

AI Governance Council: findings

Experiment: three LLM seats (Gemini 3.8 Flash via Google AI Studio, gpt-oss 120B via Groq, Ling 3.0 Flash via OpenRouter free tier) independently classify 8 synthetic AI use cases under EU AI Act risk tiers (Unacceptable / High-Risk / Limited Risk / Minimal Risk), then deliberate on disagreements. Council verdict = majority vote, ties broken by mean confidence. Run 2026-09-27. All prompts are synthetic; free tiers may use prompts for model improvement, so nothing sensitive was submitted.

Headline results

	Accuracy	95% CI
Council (majority + confidence tie-break)	7/8 (87.5%)	0.53-0.98
Gemini 3.8 Flash (round 1)	6/7 completed (85.7%)	0.49-0.97
gpt-oss 120B (round 1)	7/8 (87.5%)	0.53-0.98
Ling 3.0 Flash (round 1)	4/5 completed (80.0%)	0.38-0.96

The council matched the best individual seat (gpt-oss) and beat the other two. With n=8, every interval is wide: this is a pilot, not a benchmark.

The one miss: case 7

AI-generated product descriptions with human editorial review before publication. Ground truth assigned: Limited Risk (transparency obligation). Council verdict: Minimal Risk (1-1 tie between Gemini at 0.95 confidence and gpt-oss at 0.78, broken toward the more confident vote).

This miss deserves an asterisk: it is legally ambiguous, not a clear model error. Article 50(2) of the EU AI Act exempts AI-generated text from machine-disclosure obligations where it has undergone human review with editorial responsibility, which is exactly what the case describes. Gemini's Minimal Risk vote cited that exemption explicitly. Reasonable analysts disagree here, and the council surfaced a genuine ambiguity rather than making a simple mistake. If the ground truth is debatable, the council is arguably 8/8.

Deliberation: what actually happened

Deliberation ran on 5 of 8 cases. Across all of them, exactly **one** vote changed: Gemini on case 8, which had failed round 1 with a quota error and used the deliberation round to cast its first real vote (High-Risk, correct). No seat was ever persuaded to change a cast vote. Cases 5 and 7 ended in 1-1 ties settled by the confidence rule.

Two lessons: (1) with only 8 cases and high round-1 agreement, deliberation had little room to matter; (2) the confidence-weighted tie-breaker is double-edged. It saved case 5 (Unacceptable at 0.96 beat High-Risk at 0.88) and sank case 7 (Minimal at 0.95 beat Limited at 0.78).

Confidence carried no signal

Mean confidence on correct round-1 votes: 0.943 (n=17). Mean confidence on wrong votes: 0.940 (n=3). The models were equally sure when right and when wrong, and the most confident wrong vote in the experiment (0.95, case 7) decided the council's only miss. Any aggregation rule that trusts stated confidence, as this council's tie-breaker does, inherits that miscalibration.

Free-tier reliability: the infrastructure story

This experiment's binding constraint was not model quality but free-tier reliability. 9 of 24 seat-case round-1 slots failed in the original run (37.5%), and all 9 repair re-attempts failed again:

Seat	Failures (original)	Cause	Repairs recovered
gpt-oss 120B (Groq)	0/16 calls	-	-
Gemini 3.8 Flash (Google)	1/16 (case 8 r1)	HTTP 429 quota	0 (still 429)
Ling 3.0 Flash (OpenRouter free)	8/16	empty / non-JSON responses	0 of 8

Ling failed all 5 deliberation-round prompts in the original run and all 5 again on repair: 40 consecutive empty or unparseable responses across both attempts for the longer round-2 prompt format. Groq was flawless across all 16 calls. Practical takeaway: on free tiers, output-format compliance varies more than classification accuracy, and any multi-model design needs a retry-and-quorum policy, not just a voting rule.

Agreement

Pairwise round-1 agreement on completed votes: gpt-oss vs Ling 5/5, Gemini vs gpt-oss 5/7, Gemini vs Ling 3/4. Cases 1, 2, 3 were unanimous in round 1.

Limitations

- n=8 synthetic cases, written and labeled by one person (me). No held-out set, no independent labeling, single run.
- Ground truth for case 7 is debatable (see above).
- Free-tier failures mean Ling's individual accuracy rests on 5 completed votes, not 8.
- Confidence intervals are Wilson score; with n=8 they are wide by

construction.

Reproducibility

`cases.json` holds the 8 cases. `run\_council.py` runs the experiment (requires the provider CLI clients and stored credentials).

`repair\_votes.py` re-runs failed votes without altering recorded verdicts.

`analyze.py` reproduces every number above from `results.json`.